PROJECT UDYAM DIGITAL

Mandatory 1-Page Privacy & Security Note · Statutory Compliance under India's Digital Personal Data Protection Act (DPDP Act), 2023

Airtel Sync 2026 · Track A
DPDP ACT 2023 SHIELD

1. Legal Consent & Purpose Limitation

DPDP §6 & §7

Project Udyam Digital treats small business owners as autonomous Data Principals:

- **Itemized, Unconditional Consent Screen:** Before document ingest, users receive an explicit, bilingual notice detailing the exact processing purpose (telecom KYC, port feasibility, and statutory fraud prevention).
- **Granular Revocation Mechanism:** Consent tokens carry cryptographic identifiers (`CONSENT_YYYY_MM_DD_UUID`). If consent is declined or withdrawn, session caches are purged instantly.
- **Zero Bundled Approvals:** KYC consent is never co-mingled with marketing, ad-tracking, or third-party lead brokers.

2. Zero-Persistence In-Memory Redaction

DPDP §8

National identity numbers (Aadhaar 12-digit UIDs, PAN, Signatory voter IDs) are treated as high-risk PII:

- **RAM-Level Byte Filtering:** During OCR parsing (via PyMuPDF and Regex), raw byte buffers are parsed in transient RAM and redacted into non-numeric tokens (`[Aadhaar Redacted-XXXX]`) immediately.
- **Zero Unmasked Disk Exposure:** Neither SQLite registries, local caches, application logs, nor cloud telemetry ever persist unredacted 12-digit identity numbers.
- **Verhoeff Checksum Without Persistence:** Checksum mathematical verification occurs in-flight; raw digits are discarded immediately after validity confirmation.

3. Generative AI & Voice Privacy Firewall

DATA MINIMIZATION

Strict boundary enforcement isolates the external Multimodal LLM from sensitive business PII:

- **Pre-LLM Sanitization:** The conversational advisor filters user messages through localized regex scrubbers before invoking Gemini. GSTINs, phone numbers, email addresses, and turnover values are anonymized.
- **On-Device Voice Synthesis:** Audio turns utilize the browser Web Speech API. Voice sound waves never transit or persist on backend servers; only scrubbed text tokens reach the orchestrator.
- **Strict Pydantic Schema Guards:** Every LLM structured response is validated against `Pydantic v2` models with `extra="forbid"`, discarding prompt injections and unauthenticated tariffs.

4. Registry Lookups & Anti-Scraping Protection

SECURITY SHIELD

Protecting the 3,534 GSTIN and 2,723 Udyam merchant database from enumeration attacks:

- **Strict Exact-Match Invariant:** Registry queries require full 15-character GSTINs or 19-character Udyam numbers. Prefix searching, wildcard scans, and "did you mean" autocompletes are explicitly disabled.
- **IP Hash Telemetry:** Requesting IP addresses are hashed using one-way SHA-256 before logging to prevent user geolocation tracking.
- **Rate-Limiting & Air-Gap:** Mock government API lookups are throttled per session token, preventing bulk competitor crawling of competitor trade names or turnover bands.

5. Data Lifecycle & Retention Matrix

ENTERPRISE ARCHITECTURE

Data Category	Specific Elements Collected	Lawful Purpose & Justification	Storage Medium	Retention Horizon
Business Identity	Trade name, 15-char GSTIN, Udyam Number, Segment, City Tier	Catalog eligibility, CPQ discount calculation, statutory B2B tax compliance	FastAPI in-memory registry / Relational DB	Active contract term + 30 days post-churn
Signatory Verification	Signatory Name, masked ID token (`[ID Redacted-XXXX]`)	Legal authority confirmation and digital service agreement execution	Ephemeral RAM buffer (PyMuPDF in-flight)	Purged from memory < 2 seconds post-verification
Premises Utility Proof	Electricity bill date, address line, utility consumer reference	Physical optical fiber trenching feasibility and address freshness (<90 days)	Transient memory verification	Deleted immediately after SLA confirmation
Conversational Context	Team size, preferred Indic language, budget limit, add-on needs	Deterministic DAG bundling and optimal tariff recommendation	Session token cache (browser session)	Auto-expires on tab close or session reset

Architectural Invariant Guarantee for Hackathon Jury Evaluation:
Project Udyam Digital achieves regulatory compliance not as an afterthought or compliance checklist, but as a mathematical constraint baked into the system architecture. By enforcing zero-persistence in-memory redaction, deterministic schema validation, and air-gapped LLM boundaries, the platform ensures that 6 Crore+ Indian MSMEs can onboard digitally with zero risk of identity theft, credential leakage, or unauthorized data monetization.

Team Neural Nexus (Hrushikesh Ramilla & G Yaswanth Kumar) · ABV-IIITM Gwalior

Live Cloud Monolith: **http://65.1.112.251** · Evaluation Anchor: **2026-07-15**